BEFORE STARTING

1. Seats ..... POSITION AND LOCK
2. Seat Belts and Shoulder Harnesses ..... FASTEN AND ADJUST
3. Parking Brake ..... SET
4. All Avionics ..... OFF
5. Landing Gear Handle ..... DOWN
6. Cowl Flap Switches ..... CHECK, OPEN
7. Fuel Selectors ..... CHECK ON
8. All Circuit Breakers, Switches and Equipment Controls ..... CHECK
9. Battery Switch ..... ON
10. Beacon ..... ON
11. Fuel Quantity Indicators ..... CHECK QUANTITY
12. Landing Gear Position Lights ..... CHECK
13. Emergency Landing Gear Handcrank ..... STOWED

STARTING (Repeat for second engine)

1. Mixture Control ..... FULL RICH
2. Propeller Control ..... FULL FORWARD
3. Throttle ..... FULL OPEN
4. Fuel Boost Pump Switch ..... HI - until fuel flow peaks then OFF
5. Throttle ..... CLOSE, THEN OPEN 1/2 INCH

CAUTION

Do not engage starters for more than 30 seconds in any 4-minute period.

6. Magneto/Start Switch ..... START
7. Throttle ..... 900 to 1000RPM after start

NOTE

If the engine is hot, and the ambient temperature is 32°C or above, place mixture control in IDLE CUT-OFF, switch fuel boost pump to HI for 30 to 60 seconds, then OFF. Return mixture control to FULL RICH.

In the event of a balked start (or overprime condition) place mixture control in IDLE CUT-OFF and open the throttle; operate the starter to remove excess of fuel. As engine starts, reduce the throttle to idle rpm and place the mixture control in FULL RICH.

8. Oil Pressure ..... 10 PSI WITHIN 30 SECONDS
9. Alternator Switch ..... ON
10. Loadmeters and Voltmeter ..... CHECK FOR BATTERY CHARGE
11. Voltmeter ..... CHECK FOR 28 VOLTS
12. Red START Annunciator Light ..... CHECK OFF

AFTER STARTING AND TAXI

CAUTION

Do not operate engine above 1200 RPM until oil temperature reaches 24°C or above and oil pressure is in the green arc.

1. Avionics ..... ON, AS REQUIRED
2. Exterior Lights ..... AS REQUIRED
3. Brakes ..... RELEASE AND CHECK

BEFORE TAKEOFF

1. Parking Brake ..... SET
2. Seat Belts and Shoulder Harnesses ..... CHECK
3. Fuel Boost Pumps ..... OFF (if OAT > 32°C, use LOW pres boost)
4. All Instruments ..... CHECKED
5. Fuel Indicators ..... CHECK QUANTITY INDICATED
6. Mixture ..... FULL RICH
7. Starter Energized Annunciator Light ..... CHECK OFF
8. Throttles ..... 2200 RPM
9. Propeller ..... EXERCISE
10. Throttles ..... 1700 RPM
11. Mag CHECK (50rpm max diff. between mags/150rpm max drop)
12. Throttles ..... 1500 RPM
13. Propellers ..... CHECK FEATHERING
14. Throttles ..... IDLE (Note RPM)
15. Throttles ..... 900 to 1000 RPM
16. Electric Trim ..... CHECK OPERATION
17. Flaps ..... CHECK AND SET FOR TAKEOFF
18. Flight Controls ..... CHECK
19. Elevator Trim ..... T.O.
20. Doors and Windows ..... CLOSED AND LOCKED
21. Parking Brake ..... OFF

TAKEOFF

Landing Lights ..... ON

Transponder ..... ALT

Takeoff Power ..... Full Throttle, 2700 rpm

Minimum Takeoff Oil Temperature ..... 24°C

SET TAKEOFF POWER (MIXTURE-FULL RICH) BEFORE BRAKE

6. Airspe ACCELERATE TO AND MANTAIN RECOMMENDED SPEEDS

7. Landing Gear ..... RETRACT

8. Landing Lights ..... OFF

9. Airspeed ..... ESTABLISH DESIRED CLIMB SPEED

10. Flaps ..... RETRACT

CRUISE CLIMB

1. Mixture ..... FULL RICH
2. Cowl Flaps ..... AS REQUIRED
3. Power ..... FULL THROTTLE
4. Propellers ..... 2500 RPM
5. Engine Temperatures ..... MONITOR
6. Fuel Boost Pumps ..... OFF

FUEL FLOW SCHEDULE FOR FULL THROTTLE AND 2500 RPM			
PRESSURE ALT (FEET)	FUEL FLOW GPH/ENG	PRESSURE ALT (FEET)	FUEL FLOW GPH/ENG
SL	24.6 - 26.3	10000	17.5 - 20.0
2000	23.2 - 24.2	12000	16.3 - 19.0
4000	21.8 - 23.4	14000	14.8 - 18.0
6000	20.3 - 22.3	16000	13.4 - 17.0
8000	18.8 - 21.2	17000	12.7 - 16.5

FUEL FLOWS SHOWN ARE FOR BEST RATE OF CLIMB AND MAY VARY WITH AIRPLANE ATTITUDE

CAUTION

Engine roughness at high altitudes or fuel flow fluctuation or low fuel flow can occur when climbing on hot days. These can be eliminated by switching the fuel boost pump to LO and manually leaning to the prescribed fuel flow schedule.

MANUALLY LEANING FUEL FLOW SCHEDULE FOR FULL THROTTLE AND 2500 RPM WITH LOW BOOST			
PRESSURE ALT (FEET)	FUEL FLOW GPH/ENG	PRESSURE ALT (FEET)	FUEL FLOW GPH/ENG
SL	25.5	10000	18.8
2000	23.7	12000	17.7
4000	22.6	14000	16.4
6000	21.3	16000	15.2
8000	20.0	17000	14.6

NOTE

After adjustment of the mixture control, setting the fuel flow per the manual leaning schedule, the Engine Driven Fuel Pump will automatically lean engine mixture for the airplane's pressure altitude as the airplane climbs.

CAUTION

Return the mixture control to FULL RICH before switching the fuel boost pump to OFF.

CRUISE

See Cruise Tables and MANIFOLD PRESSURE vs RPM graph in PERFORMANCE Section.

1. Cowl Flaps ..... CLOSED
2. Power ..... SET
3. Fuel Boost Pumps ..... AS REQUIRED
4. Mixtures ..... SET USING EGT

DESCENT

1. Altimeter ..... SET
2. Mixture ..... FULL SET
3. Cowl Flaps ..... CLOSED
4. Flaps ..... AS REQUIRED
5. Power ..... AS REQUIRED

Avoid prolonged idle settings. Cylinder Head Temperatures are not to fall below 149°C for longer than five minutes.

RECOMMENDED DESCENT SPEEDS

16,000 to 13,000 ft ..... 160 kts
Below 13,000 ft ..... 170 kts

LANDING

1. Seat Belts and Shoulder Harnesses ..... FASTENED
2. Fuel Selector Valves ..... SELECT FULLER TANK
3. Fuel Boost Pumps ..... OFF, OR LOW AS PER OAT
4. Cowl Flaps ..... AS REQUIRED
5. Mixture Controls ..... FULL RICH
6. Flaps ..... APPROACH (15°) below 152 kts
7. Landing Gear ..... DOWN below 152 kts
8. Landing Lights ..... ON
9. Flaps ..... FULL DOWN below 122 kts
10. Airspeed ..... ESTABLISH NORMAL LANDING APPROACH SPEED
11. Propellers ..... HIGH RPM

BALKED LANDING

1. Propellers ..... HIGH RPM
2. Power ..... MAXIMUM ALLOWABLE
3. Airspeed ..... 95 kts
4. Flaps ..... UP
5. Landing Gear ..... UP
6. Cowl Flaps ..... AS REQUIRED

AFTER LANDING

1. Landing and Taxi Lights ..... AS REQUIRED
2. Flaps ..... UP
3. Trim Tabs ..... SET TO ZERO
4. Cowl Flaps ..... OPEN
5. Fuel Boost Pumps ..... AS REQUIRED

SHUTDOWN

1. Parking Brake ..... SET
2. Propellers ..... HIGH RPM
3. Throttles ..... 1000 RPM
4. Fuel Boost Pumps ..... OFF
5. Electrical and Avionics Equipment ..... OFF
6. Mixture Controls ..... IDLE CUT-OFF
7. Magneto/Start Switches ..... OFF, AFTER ENGINES STOP
8. Battery and Alternator Switches ..... OFF